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InfoSec Lab-03

⇒ Exp 1:

Q1: text = MEET ME AT THE TOWER

Total Letters = 16

Letter	Count	Frequency (%)
M	2	12.5 %
E	5	31.25 %
T	4	25 %
A	1	6.25 %
H	1	6.25 %
O	1	6.25 %
W	1	6.25 %
R	1	6.25 %

Q2: text = ATTACK AT DAWN

Total Letters = 12

Letter	Count	Frequency
A	4	33.3 %
T	3	25.0 %
C	1	8.3 %
K	1	8.3 %
D	1	8.3 %
W	1	8.3 %
N	1	8.3 %

Most Frequent Letter = 'A'

Its Frequency = 33.3 %

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Q3: Spaces & punctuation is ignored in frequency analysis because they do not contribute in letter distribution. These are not part of language's natural structure.

Q4: A 10-letter message is too small to reflect true language patterns, so its letter frequencies vary widely by chance while a 1000-letter message is large enough for those patterns to stabilize & reliably match real language statistics.

⇒ Exp 2:

∴ Key = (cipher-letter − 'E') mod 26 ; [A-Z → 0-25]

Q1: Most Freq. Letter = H → 8

Key = (8 − 4) % 26 = 4

Key = 4

Q2: Most Freq Letter = B → 1

Key = (1 − 4) % 26 = −3 % 26

Key = 23

Q3: cipher = VNNC VN JC CQN URKAJAH

Total letters = 18

letter	count	frequency %
V	2	11.1 %
N	4	22.2 % ← Most Freq
C	3	16.6 %
J	2	11.1 %
Q	1	5.5 %
U	1	5.5 %
R	1	5.5 %
K	1	5.5 %
A	2	11.1 %
H	1	5.5 %

⇒ N → 14

Key = 13 - 4

Key = 9

V → 21-9 = 12 → (M)

N → 13-9 = 4 → (E)

C → 2-9 = -7 %26 = 19 → (T)

J → 9-9 = 0 → (A)

Q → 16-9 = 7 → (H)

U → 20-9 = 11 → (L)

R → 17-9 = 8 → (I)

K → 10-9 = 1 → (B)

A → 0-9 = -9 %26 = 17 → (R)

H → 7-9 = -2 %26 = 24 → (Y)

Plain Text = "MEET ME AT THE LIBRARY"

Q4: cipher = WII XLI XIEGLIV XSQSVVSA

Total letters = 21

letter	count	
W	1	
I	5	←
X	3	
L	2	
E	1	
G	1	
V	3	
S	3	
Q	1	
A	1	

W → 22-4 = 18 → (S)

I → 8-4 = 4 → (E)

X → 23-4 = 19 → (T)

L → 11-4 = 7 → (H)

E → 4-4 = 0 → (A)

G → 6-4 = 2 → (C)

V → 21-4 = 17 → (R)

S → 18-4 = 14 → (O)

Q → 16-4 = 12 → (M)

A → 0-4 = -4 % 26 = 22 → (W)

∴ Let I = E (4)

I → 8

Key = 8-4

Key = 4

P = SEE THE TEACHER TOMORROW

Q5: Vignere cipher can't easily be broken by frequency analysis because it uses a repeating keyword to change the cipher alphabet for every letter. This polyalphabetic encryption flattens and masks normal letter frequencies (like 'E' or 'T'), making the cipher look completely random until the exact lenght of the key is discovered.

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⇒ Exp 3:-

Q1: P = SEE YOU SOON

Plain:	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Cipher:	Q	W	E	R	T	Y	U	I	O	P	A	S	D	F	G	H	J	K	L	Z	X	C	V	B	N	M

S → L
E → T
Y → N
O → G
U → X
S → L
N → F

Cipher = LTT NGX LGGF

Q2: cipher = LTT NGX

L → S
T → E
N → Y
G → O
X → U

P = SEE YOU

Q3: X (14 times), Q (9 times), M (8 times)

- Using the English frequency order: ETAOINSHRDLU

X → E
Q → T
M → A

Q4: Brute force is impractical because a general substitution cipher has 26! possible keys, approx. 4×10^{26}. Frequency analysis works by comparing the frequencies of ciphertext letters with common English letter patterns & using repeated words to infer the substitutions.

⇒ Exp 4:-

Q1: cipher = DWWDFN for keys 0, 1, 2 and 3.

Key	Decrypted Result
0	DWWDFN
1	CVVCEM
2	BUUBDL
3	ATTACK ✓

Q2: chr(((ord(B) − 65 − 5) mod 26) + 65)

Steps:

1) ord('B') = 66
2) 66 − 65 = 1
3) 1 − 5 = −4
4) −4 mod 26 = 22
5) 22 + 65 = 87
6) chr(87) = 'W'

Ans = 'W'

Q3:

- The outer loop runs 26 times, once for each possible key.
- Each iteration decrypts 6 letters.

$$26 \times 6 = 156$$

Therefore, the total is 156 letter decryptions.

Q4: The space would be omitted because the program would no longer append non-alphabetic characters to the decrypted string. The output for the correct key would become 'HELLOWORLD' instead of HELLO WORLD.

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